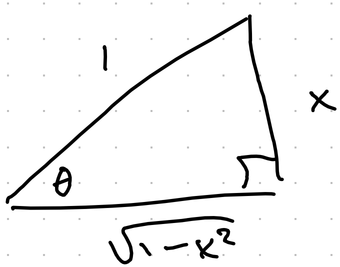
Derivatives of inverse trig functions − 0.3.7

Computing the derivative of arcsin(x):

$f(x) = \sin(x) \Rightarrow f'(x) = \cos(x)$

$f^{-1}(x) = \arcsin(x)$

$$\frac{d}{dx}\arcsin(x) = \frac{1}{f'(f^{-1}(x))} = \frac{1}{\cos(\arcsin(x))} = \frac{1}{\cos\theta}$$



Pythagorean thm: $x^2 + y^2 = 1$

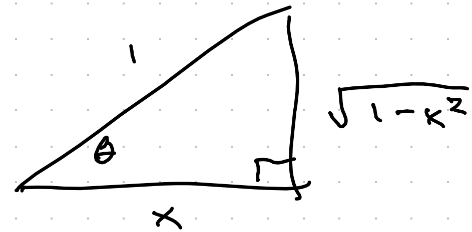
$y^2 = 1 - x^2 \Rightarrow y = \sqrt{1-x^2}$

$\cos\theta = \sqrt{1-x^2}$

$$\Rightarrow \boxed{\frac{d}{dx}\arcsin(x) = \frac{1}{\sqrt{1-x^2}}}$$

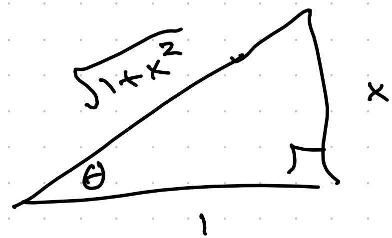
$$\frac{d}{dx}\arccos(x) = \frac{1}{-\sin(\arccos(x))}$$

$$= \frac{-1}{\sin\theta} = \frac{-1}{\sqrt{1-x^2}}$$



$$\frac{d}{dx}\arctan(x) = \frac{1}{\sec^2(\arctan(x))}$$

$$= \frac{1}{\sec^2\theta} = \frac{1}{1+x^2}.$$



$$\sec\theta = \frac{\text{hyp}}{\text{adj}} = \sqrt{1+x^2}$$

$$\sec^2\theta = 1+x^2$$

Example: Find $\frac{d}{dx} \arccos(e^x + 1)$.

$$\frac{d}{dx} \arccos(e^x+1) = \frac{-1}{\sqrt{1-(e^x+1)^2}} \cdot e^x = \frac{-e^x}{\sqrt{1-(e^x+1)^2}}.$$

Example: Find $\frac{d}{dx} \ln|\arcsin(x)|$.

$$\frac{d}{dx} \ln|\arcsin(x)| = \frac{1}{\arcsin(x)} \cdot \frac{1}{\sqrt{1-x^2}}.$$